



Inspiring Excellence

CSE370 : Database Systems
Project Report
Project Title : Study_Group_Finder

Group No : _12_, CSE370 Lab Section : _10_, Spring 2026		
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Introduction

The **Study Group Finder System** is a web-based platform designed to enhance academic collaboration among university students by helping them easily find, create, and manage study groups. In many academic environments, students often struggle to connect with classmates who share the same courses, schedules, or academic interests. This system addresses that challenge by providing a structured and efficient way to bring students together for collaborative learning.

The platform allows students to register using their academic details such as name, student ID, department, semester, and email. Based on this information, the system organizes users and recommends relevant study groups. Students can create course-based study groups, join existing groups, or send requests to join private groups, which are managed by group administrators.

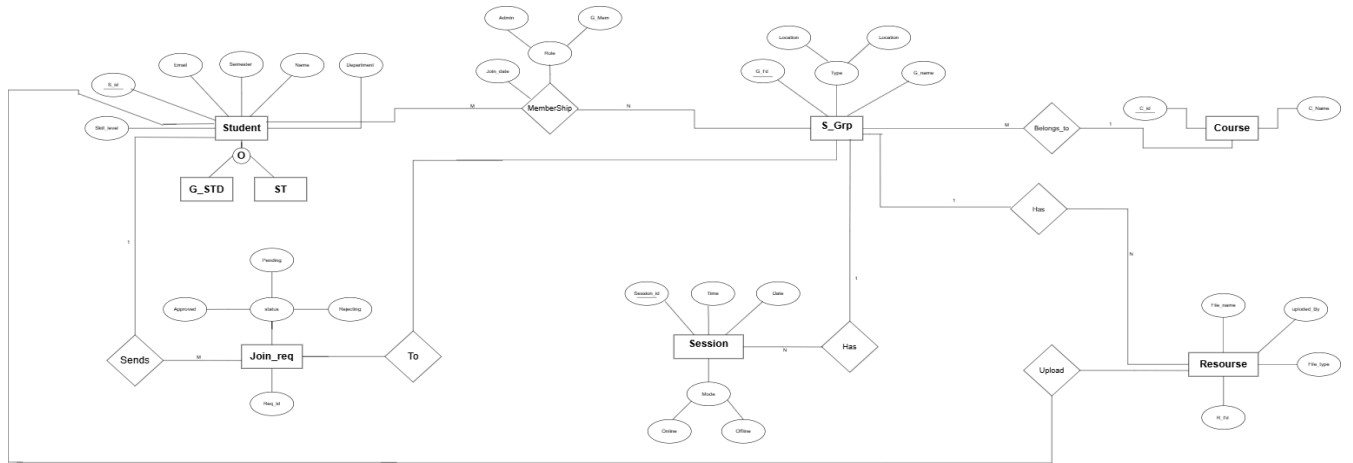
To further support effective learning, the system includes features such as study session scheduling, skill-level matching (beginner, intermediate, advanced), resource sharing, and study partner matching. Additionally, it provides smart recommendations based on shared courses, similar schedules, and preferred study locations such as libraries, cafeterias, study rooms, or online platforms.

By combining these features, the Study Group Finder System aims to improve student productivity, encourage teamwork, and create a more connected academic environment where learning becomes more interactive and accessible.

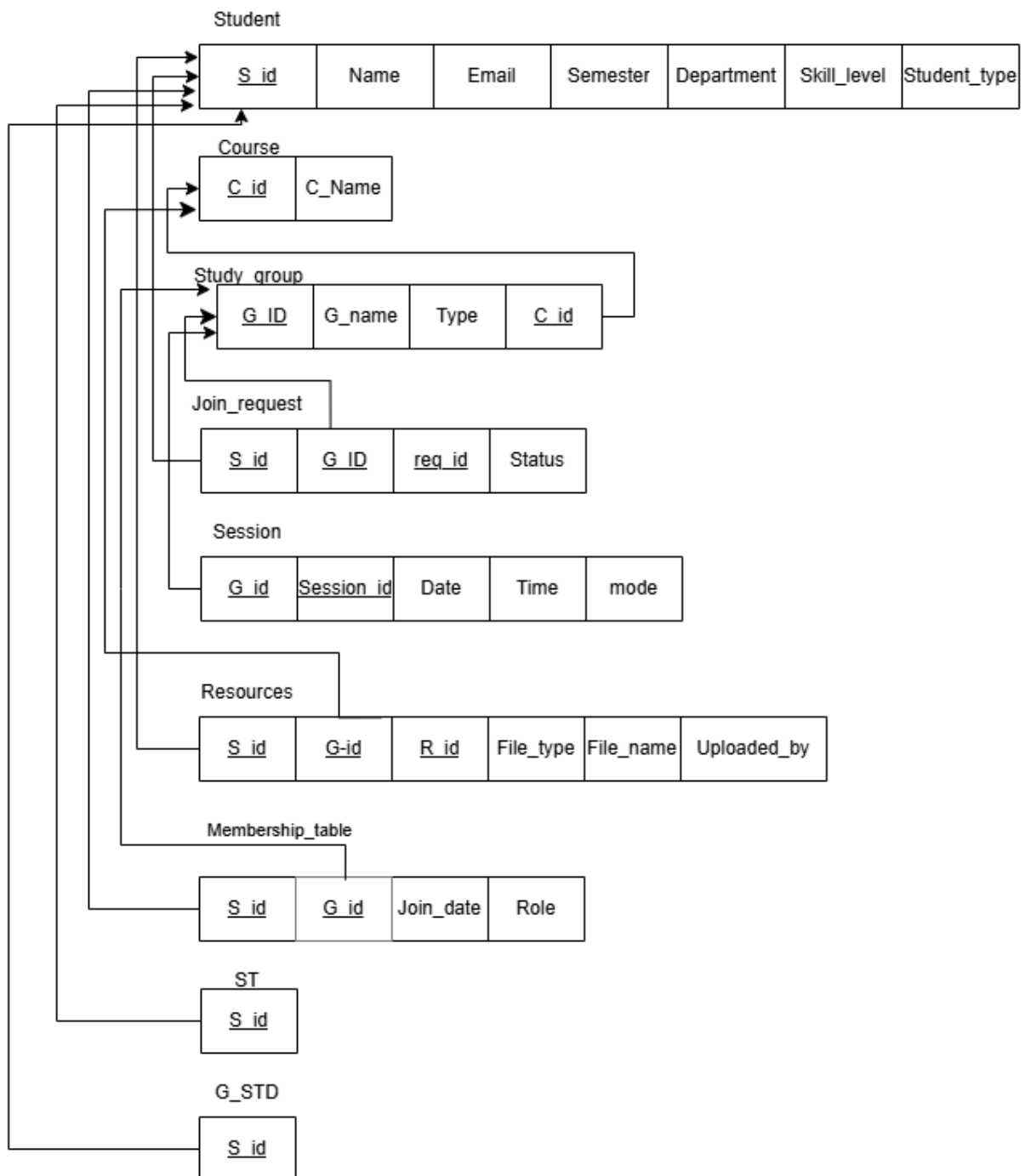
Project Features

ID, Name	Features [2 per member]	
22201923, Abdullah Al Noor	Ft 1	Resource verification
	Ft 2	Resource upload and download.
22301762, Md. siddique bishwas	Ft 1	Group mate matching.
	Ft2	Session finder.
24201463, Md.Tanimur Rahman	Ft 1	Implemented group creation functionality.
	Ft 2	join request system for users.

ER/EER Diagram

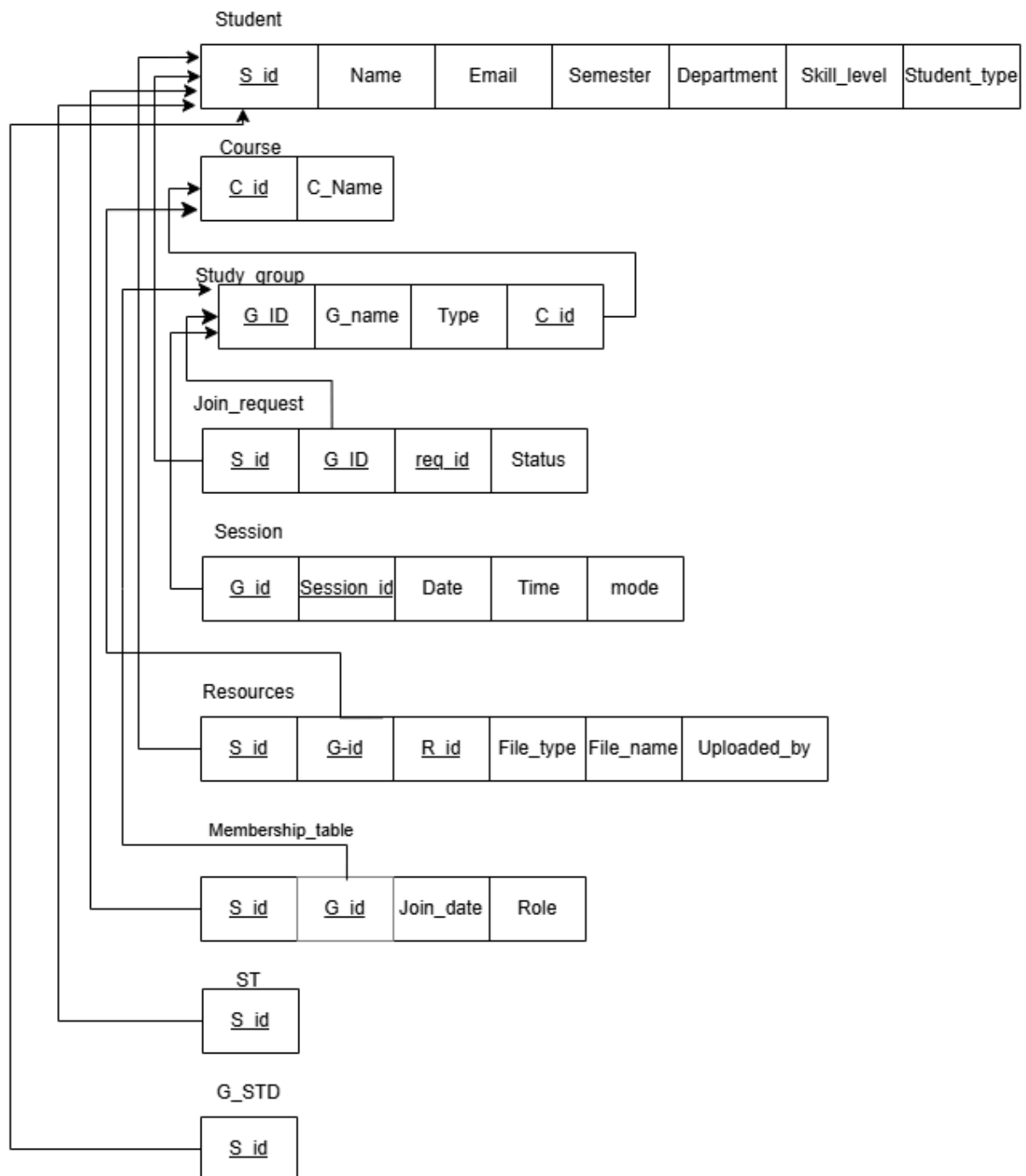


Schema Diagram



Normalization

- a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
 - b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
 - c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?
-
- a. Yes this is converted to 1NF. Because there are no multivalued/composite attributes.
 - b. Yes this is converted to 2NF. Because there is no partial relationship with the key.
 - c. Yes this is converted to 3NF. Because there is no transitive relation here.



Frontend Development

Contribution of ID : 22201923, Abdullah Al Noor

File Verification & Status Indicators

Verification results are communicated through visual cues to ensure users only interact with valid, uncorrupted data.

- **Status Badges:** Next to each file in the list, a status indicator (often a green checkmark or a red warning icon) reflects the results from the `verify_files.php` script.
- **Error Messaging:** If a file fails verification or a download is interrupted, the frontend displays clear alerts using CSS-styled boxes to explain the issue (e.g., "File Not Found" or "Access Denied").
- **Validation Feedback:** When a user attempts to upload an invalid file, JavaScript provides instant feedback, preventing unnecessary server requests and improving the overall experience.

File Download & Resource Lists

The download interface is integrated into the course and group management views, organizing shared content into scannable lists.

- **Dynamic Tables:** Within `view_course.php`, files are displayed in a tabular format showing the filename, size, and the date it was uploaded.
- **Secure Links:** Instead of direct links to server folders, the frontend uses buttons that point to `download_file.php?id=....`. This masks the actual file location on the server, enhancing security.
- **User Permissions:** The UI dynamically hides or shows download buttons based on the user's session status, which is verified by the `auth.php` script.

File Upload Interface

The upload module is designed to be accessible directly from the user dashboard and group pages to minimize navigation.

- **Interactive Forms:** The dashboard.php and groups.php views feature HTML `<form>` elements with the `enctype="multipart/form-data"` attribute, which is essential for transmitting binary file data to the server.
- **File Selection:** A stylized input field allows users to browse their local directory. JavaScript is used to restrict selection to allowed file types (e.g., .pdf, .docx, .zip) before the upload even begins.
- **Action Buttons:** Dedicated "Upload" buttons trigger the `upload_file.php` backend script, often accompanied by a progress bar or loading spinner to indicate the transfer status.

Contribution of ID : 22301762, Md. siddique bishwas

In this project, I developed the frontend for the matching and session creation features using HTML and CSS. I designed a user interface where users can create study sessions by filling out a form with details such as subject, time, and preferences using elements like `<input>`, `<select>`, and `<button>`.

Contribution of ID : 24201463, Name : Md. Tanimur Rahman

In this project, I developed the frontend for the group creation and join request features using HTML and CSS. I designed a form where users can create a study group by providing information such as group name, course, and description using input fields and form elements like `<input>`, `<textarea>`, and `<button>`.

I also created a structured layout to display available groups dynamically, where each group includes a “Join Request” button. This button is implemented using HTML forms and is connected to backend PHP scripts through methods like `POST` to send user requests.

Additionally, I worked on styling the interface using CSS to make it user-friendly and responsive, ensuring proper alignment, spacing, and layout. Overall, my frontend ensures smooth interaction between the user and the backend system.

Backend Development

Contribution of ID : 22201923, Name : Abdullah Al Noor

File Verification System

The verification module acts as a quality control layer, ensuring that all hosted files are valid and reach the intended recipients without corruption.

- **Integrity Checks:** `upload_handler.php` performs routine checks to confirm that the file path recorded in the database matches a physical file on the server disk.
- **Validation Logic:** The system utilizes backend functions to verify file types and checksums, preventing users from spoofing file extensions (.exe or any application file can't be uploaded by the user).
- **Status Reporting:** Verification results are often reflected in the `view_course.php` or `groups.php` interfaces, providing visual feedback to users regarding the availability of their documents.

File Upload Implementation

The upload process converts user-selected local files into server-side assets while maintaining data integrity.

- **Request Handling:** `upload.php` serves as the entry point, processing **POST** requests containing file data.
- **Directory Management:** Uploaded files are moved from temporary storage to a designated project directory to ensure they are accessible for future retrieval.
- **Database Registration:** Metadata for each file—including the original filename, file path, and the ID of the uploading user—is recorded in the database defined by `database.sql`.
- **Security Measures:** Logic within `includes/functions.php` checks for file size limits and allows extensions to prevent the execution of malicious scripts.

File Download Mechanism

The download system allows users to retrieve stored documents while keeping the actual file paths hidden from the public URL.

- **Retrieval Logic:** download_file.php queries the database to locate the physical file associated with a specific resource ID.
- **Header Manipulation:** The script sends specific HTTP headers to the browser to trigger a "Save As" dialogue, ensuring the file is downloaded rather than rendered in the browser window.
- **Access Control:** Integration with auth.php ensures that only authorized users or members of a specific group can download sensitive project files.

Contribution of ID : 22301762, Name :Md. Siddique Bishwas

The matcher.php code is the Study Partner Matcher module.

1. Requires login – only authenticated users can access it.
2. Auto-filters potential partners by the current user's department and skill level (or manual filter via GET form).
3. Queries the Users table with prepared statements to find matching students (excluding the logged-in user, max 30 results).
4. Displays each match as a card with a "Match" button.
5. Handles POST requests to insert a new record into the S_Partner table, after verifying:
 - CSRF token
 - Valid partner ID (not self)
 - No duplicate match exists
6. Uses output escaping (e()) and prepared statements throughout, preventing XSS and SQL injection.

Overall, it's a secure, simple tool for finding study partners with similar academic backgrounds.

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Contribution of ID : 24201463, Name : Md. Tanimur Rahman

Group Creation

The group creation feature allows users to establish new collaborative spaces within the platform.

- **Implementation Logic:** Handled by The matcher.php code is the Study Partner Matcher module. It:
-
- **Requires login** – only authenticated users can access it.
-
- **Auto-filters potential partners** by the current user's department and skill level (or manual filter via GET form).
-
- **Queries the Users table** with prepared statements to find matching students (excluding the logged-in user, max 30 results).
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- **Displays each match** as a card with a "Match" button.
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- **Handles POST requests** to insert a new record into the S_Partner table, after verifying:
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- **CSRF token**
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- **No duplicate match** exists
-
- **Uses output escaping (e())** and prepared statements throughout, preventing XSS and SQL injection.
-
- **Overall, it's a secure, simple tool** for finding study partners with similar academic backgrounds.
-
- **, the system processes user inputs** to generate new records in the database.
- **Database Integration:** The script interacts with the group schema defined in database.sql to store group metadata, such as names and descriptions.
- **User Dashboard:** The feature is integrated into the primary user interface via dashboard/dashboard.php, allowing for immediate access to creation tools.

Joining Groups

This feature manages the process of users discovering and becoming members of existing groups.

- **Join Requests:** The `join_group.php` script manages the logic for users to find and request entry into specific groups.
- **Request Processing:** Backend scripts like `process_request.php` and `manage_requests.php` handle the logic for approving or managing membership applications.
- **Membership Mapping:** The system maintains a relationship between users and groups, ensuring that the `view_course.php` and `groups.php` views display correct participant data.

Source Code Repository

<https://github.com/Tanim491/CSE370-Project->

Conclusion

In this project, we were able to build a Study Group Finder system that allows users to create groups and send join requests easily. While working on this project, I learned how to design frontend interfaces using HTML and CSS and how they connect with backend PHP and database operations.

Working on the group creation and join request part helped me understand how forms, buttons, and user interactions actually work in a real web application. I also faced some challenges during implementation, but solving them improved my problem-solving skills.

Overall, this project gave me practical experience and a better understanding of full-stack web development, which will be helpful for future projects.

References

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